

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed, developed, and maintained **reliable and scalable data pipelines** using **Python** and **SQL** across **PostgreSQL, MySQL, and Oracle** databases, automating ETL workflows for enterprise Java/.NET applications and reducing manual data processing by **40%**.
- Architected **Extract, Transform, and Load (ETL)** infrastructure and **dimensional data models** for enterprise business systems integration, enabling **sub-second query latency** on **200M+ daily transaction records** and supporting cross-functional analytics requirements for banking and payments operations.
- Partnered closely with **cross-functional stakeholders** including business analysts, operations teams, and banking-domain partners to translate evolving analytics and payments-processing requirements into actionable technical data solutions, delivering **15+ production reports** with **99.5% SLA uptime**.
- Transitioned investigative prototypes and operational metrics into **stable, production-grade reporting environments** using **Power BI** and **Tableau**, designing interactive dashboards for business intelligence and data visualization that reduced investigative turnaround time by **35%** across payments operations teams.
- Advocated **data quality** through authoring technical design documents, executing **code reviews** on data pipeline changes, and proactively resolving complex production bugs in **distributed data systems**; implemented automated data validation checks using **Python** scripts that caught **98% of schema anomalies** before downstream consumption.
- Optimized **data infrastructure and data platform** performance by refactoring legacy SQL queries and **data models**, achieving **45% reduction** in query execution time and reducing storage costs by **\$120K annually** while maintaining single source of truth integrity across interconnected ERP and enterprise systems.

SKILLS

Data Engineering & Pipelines – ETL/ELT architecture design, data pipeline development, Extract Transform Load, Extract Load Transform, distributed data systems, data infrastructure optimization, Apache Spark, dimensional data modeling.

Programming & Scripting – Python, Java, SQL, advanced SQL, shell scripting, data scripting, code reviews.

Databases & Data Modeling – PostgreSQL, MySQL, Oracle, SQL querying, schema design, data models, data warehousing, NoSQL databases, exploratory queries.

Data Visualization & BI – Power BI, Tableau, dashboard development, DAX measures, Power Query, interactive reporting, business intelligence, data visualization.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD (Jenkins, Git), Linux, distributed data processing.

Data Science & Analytics – pandas, NumPy, scikit-learn, statistical analysis, data modeling, A/B testing, time-series analysis, machine learning fundamentals.

Domain Expertise – SWIFT messaging (MT/MX), ISO 20022 message flows, SEPA payments, cross-border payments, PSD2, Open Banking, payments processing, regulatory compliance (GDPR), ERP systems integration, change-management, stakeholder management.

EDUCATION AND TRAINING

PROJECTS

Global Payments Data Lake & Real-Time Analytics Platform (HCL Technologies)

- Designed and implemented a **large-scale distributed data pipeline** processing **500M+ ISO 20022 payment messages daily** across **SEPA, cross-border, and instant payment flows** using **Python, Apache Spark**, and **advanced SQL** on **PostgreSQL** data warehouses, delivering **single source of truth** for global payments operations.
- Built **dimensional data models** for synchronous and asynchronous system integration between legacy SWIFT messaging systems and modern ISO 20022 infrastructure, enabling **99.99% transaction-flow visibility** and reducing payment reconciliation cycles from **4 hours to 12 minutes**.
- Transitioned investigative prototypes into **stable, production-grade reporting environments** with **Power BI dashboards** and **Tableau visualizations**, providing real-time operational alerting and regulatory audit trails for **12 banking partners** across **8 countries**.
- Advocated **data quality** by authoring comprehensive technical design documents, implementing **automated data validation** checks in Python that flagged schema anomalies and payment-flow inconsistencies with **99.7% precision**, and conducting rigorous **code reviews** on all pipeline changes.
- Optimized **data infrastructure performance** through refactoring **SQL queries** and **data models**, reducing query latency by **52%** on **200M+ historical transaction records** and cutting cloud storage costs by **\$180K annually** while maintaining compliance with GDPR and financial regulatory requirements.

Enterprise ERP Data Integration & Business Intelligence Hub (HCL Technologies)

- Architected **reliable and scalable ETL pipelines** integrating **3 enterprise ERP systems** and operational platforms using **Python, REST APIs**, and **SQL** to consolidate **50M+ daily records** from **PostgreSQL, MySQL, and Oracle** sources into a unified **data warehouse**.
- Developed **interactive Power BI dashboards** and **Tableau visualizations** translating cross-functional stakeholder requirements into actionable business intelligence, reducing manual reporting effort by **60%** and enabling **real-time financial and operational visibility** for **200+ business users**.
- Partnered with business analysts and operations teams through change-management workflows to validate **data quality**, execute **code reviews** on integration changes, and proactively resolve production bugs affecting **downstream analytics and reporting systems**, achieving **98.5% uptime**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)